

Frontend Open-Source Programs Playbook

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Strategy in one screen

Pick ONE primary org that runs in several programs, get 3–5 merged frontend PRs there by February 2027, and apply to every program it joins. One body of work can then count for GSoC, LFX, Outreachy and C4GT.

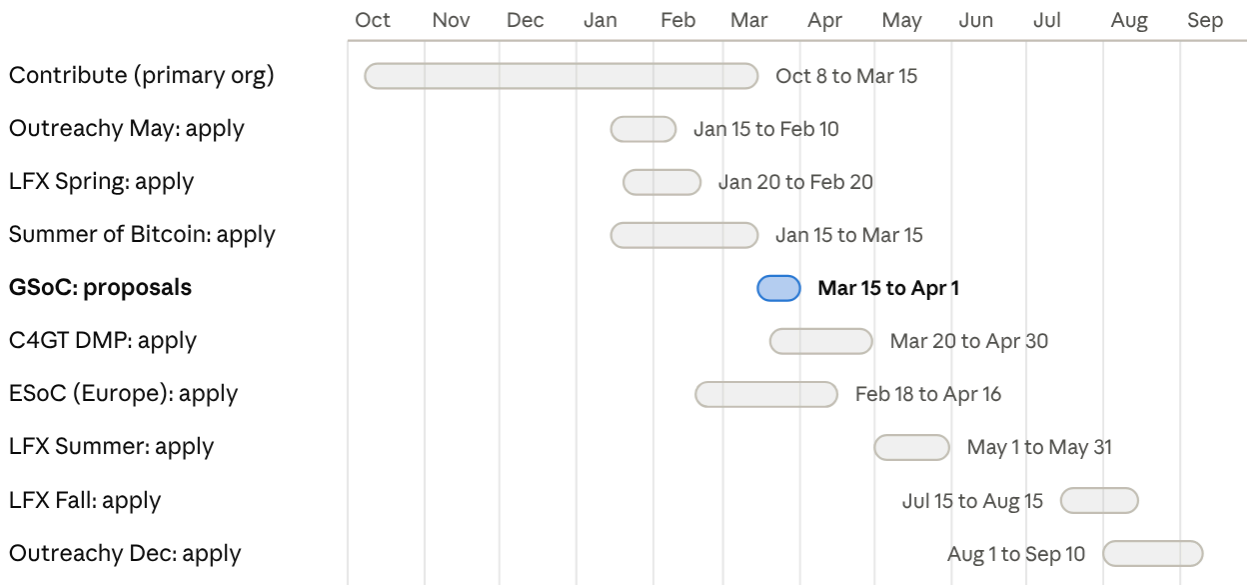
- **Primary target:** Layer5 / Meshery (React + Next.js). *Programs: GSoC, LFX, Outreachy.* One set of contributions works for all three.
- **Backup target:** one org in your second stack. CircuitVerse (Vue; GSoC) or Oppia (Angular; GSoC).
- **Low-competition track:** a C4GT DMP org. Glific, Avni or DIGIT (React; C4GT), India-focused and less crowded.
- **Stretch:** Zulip (GSoC, Outreachy) or Rocket.Chat (GSoC), once you have 2+ merged PRs elsewhere.
- **Rule of thumb:** depth in one codebase beats one PR each in ten. Mentors pick people they already know from issues and PRs.

Dates and stipends below are approximate, based on past cycles. Confirm each on the program's official site before acting on it.

Timeline

Contribute now; most application windows open Jan to Apr 2027

Approximate windows from past cycles, Oct 2026 to Sep 2027



MLH Fellowship runs rolling batches all year. Confirm every date on the official site.

application windows by program, Oct 2026 – Sep 2027

October to February is for contributing. January to April 2027 is when most applications are due, and GSoC proposals (mid-March to early April) are the main deadline.

Programs

GSoC and LFX have the most frontend slots. Outreachy and C4GT are the easiest to stand out in. Summer of Bitcoin and MLH are side bets.

Program	Who can apply	Stipend (approx.)	Application window (approx.)	Frontend scope	How selection works
GSoC	18+, students and open-source beginners	~\$750–\$3,300, scaled by country and project size	Orgs announced ~late Feb; proposals ~mid-Mar to early Apr	High: dashboards, design systems, editors, accessibility, React/Vue/Angular rewrites	Prior PRs + a proposal reviewed by org mentors

Program	Who can apply	Stipend (approx.)	Application window (approx.)	Frontend scope	How selection works
LFX Mentorship	18+, anyone	~\$3,000–\$6,600 by location	3 terms: Spring (apply ~Jan–Feb), Summer (~May), Fall (~Jul–Aug)	Medium–high: CNCF UIs (Meshery, Headlamp, Backstage plugins, KubeVela)	Cover letter + resume + a test task; prior PRs help a lot
C4GT DMP	Students and professionals, India-focused	Paid during DMP (amount varies by year)	Tickets published ~Mar–Apr; program ~May–Aug	Medium–high: React, React Native and Angular tickets on govtech products	Apply per ticket; proposal + prior contributions
Outreachy	People underrepresented in tech; initial essay required	\$7,000	2 rounds: May cohort (apply ~Jan–Feb), Dec cohort (apply ~Aug–Sep)	Medium: web projects at Wikimedia, Mozilla, GNOME, Fedora and others	Essay approval, then a ~4-week contribution period; contributions decide it
Summer of Bitcoin	University students	Paid (~\$3,000 in past years)	2026 applications closed Feb 15; expect ~Jan–Feb 2027	Low–medium: wallet and Nostr app UIs; a separate designer track (6 of 50 places in 2026)	Bitcoin + coding challenge rounds, then a proposal to one of ~26 orgs
MLH Fellowship	Students and early-career developers	Stipend in some tracks	Rolling batches every few months, 12 weeks each	Medium: Open Source and Software Engineering tracks; past partners include React, Jest and DEV	Application + technical interview; MLH matches you to a project, you don't pick the org
ESoC: European Summer of Code	Open-source newcomers worldwide	Stipend (amount varies by project)	2026 batches ~Feb–Apr; check esoc.dev for 2027	Low: mostly Python, ML and data libraries; frontend only via docs sites or dashboards	Apply per project; regional hubs in Europe do the matching

Org directory

React/TypeScript orgs have the most openings. Vue and Angular orgs get fewer applicants per slot. Program participation changes every year, so check the current org list before committing.

Org	Stack	Programs (past years)	Frontend scope	Competition
Layer5 / Meshery	React, Next.js, MUI, Sistent design system	GSoC, LFX, Outreachy	Meshery UI, Kanvas designer, Sistent components, docs site	Medium
Zulip	TypeScript, Handlebars; React Native (mobile)	GSoC, Outreachy	Web app UI, settings, accessibility, mobile app	Very high
Rocket.Chat	React, TypeScript, Fuselage UI	GSoC	Chat UI, admin panels, apps engine UI	High
Mattermost	React, TypeScript, Redux	GSoC (some years)	Web app features, plugins	High
Joplin	React, TypeScript, Electron, React Native	GSoC	Desktop/mobile note-app UI, editor	Medium-high
Wikimedia	Vue 3 (Codex design system), vanilla JS	GSoC, Outreachy	Codex components, tool UIs, accessibility	Medium
Apache Superset	React, TypeScript	GSoC (via ASF)	Charts, dashboard builder	High
Jenkins	JavaScript, React (plugins)	GSoC, LFX (CDF)	Plugin UIs, design-library migration	Medium
Backstage	React, TypeScript, MUI	LFX (CNCF)	Plugins, core UI	Medium
Headlamp	React, TypeScript	LFX (CNCF)	Kubernetes dashboard, plugins	Low-medium
CircuitVerse	Vue 3, TypeScript; Rails	GSoC	Simulator UI rewrite, Vue frontend	Medium
Oppia	Angular, TypeScript	GSoC	Lesson player, creator tools, accessibility	High

Org	Stack	Programs (past years)	Frontend scope	Competition
OpenMRS	React (O3 micro-frontends), Carbon	GSoC	O3 patient-chart apps	Medium
FOSSology	React (new UI); PHP Symfony (old UI); REST API	GSoC	New React UI (FOSSologyUI): feature parity with the old UI, REST API integration	Low-medium
AOSSIE	React, Flutter	GSoC	Many small apps, new projects each year	Medium
Sugar Labs	JavaScript, TypeScript	GSoC	Music Blocks, activity UIs	Low-medium
p5.js / Processing	React, JavaScript	GSoC	p5.js web editor, accessibility	Medium
Glific	React, TypeScript, Apollo	C4GT	NGO chatbot dashboard	Low-medium
Avni	React, React Native	C4GT	Web app, field app UI	Low-medium
DIGIT (eGov)	React micro-frontends	C4GT	Government service UIs	Low-medium
Mempool.space	Angular, TypeScript	Summer of Bitcoin	Block explorer UI, charts	Medium
BTCPay Server	C#/Razor, JavaScript	Summer of Bitcoin	Merchant dashboard, checkout	Medium
Mozilla	React, JavaScript	Outreachy	Varies by round	Medium

By stack:

- **React/Next.js:** Layer5 (GSoC, LFX, Outreachy) → Headlamp, Backstage (LFX) → Glific (C4GT) → FOSSology (GSoC) → Rocket.Chat, Joplin (GSoC)
- **Vue:** CircuitVerse (GSoC) → Wikimedia Codex (GSoC, Outreachy)
- **Angular:** Oppia (GSoC) → Mempool.space (Summer of Bitcoin)

- **Flutter/React Native:** AOSSIE (GSoC) → Avni (C4GT) → Zulip mobile (GSoC, Outreachy)
- **MLH Fellowship:** matches you to partner projects instead of you picking an org. Any of the orgs above count as experience.

Projects by org

The 2027 idea lists aren't out yet: most orgs publish them between January and February. The 2026 projects below show what each org's mentors fund, so use them to pick the codebase areas to learn now.

Org	Program	Recent projects (2025–2026)	Ideas page
Layer5 / Meshery	GSoC, LFX	2026: docs site migration from Jekyll to Hugo. 2025: Playwright UI tests (LFX), websocket and GraphQL subscription refactor, OPA policy evaluation in the UI via WebAssembly	GSoC 2026 · LFX 2026
Zulip	GSoC	2026: clusters of 3–5 full-stack features (175 or 350 hours). Wants TypeScript, HTML/CSS, visual design and Flutter skills. No small projects.	Zulip GSoC ideas
CircuitVerse	GSoC	2026: Vue Simulator integration, structured format for saved circuits, client-side Verilog synthesis, Organization dashboard with role-based access	GSoC 2026 posts

Org	Program	Recent projects (2025–2026)	Ideas page
Joplin	GSoC	2026: AI search for notes, chat with your notes, auto-categorisation, password strength indicator	Welcome to GSoC 2026
OpenMRS	GSoC	2026 candidates: CDS support in O3, program enrolment in O3, Orders and Lab app improvements	Summer of Code 2026
FOSSology	GSoC	2021: rebuild the PHP Symfony UI in React (FOSSologyUI). 2022: feature parity with the old UI, webpack to Vite migration, REST API fixes. 2026 list not found	GSoC 2022 org.page · FOSSologyUI repo · Good first issues
Rocket.Chat	GSoC	Ideas are posted in their forum; I didn't find the 2026 list	GSoC forum
Oppia	GSoC	One wiki page per year, published January–February; I didn't find the 2026 list	GSoC 2023 page (format)
Wikimedia	Outreachy, GSoC	Each project needs 2 mentors and Phabricator microtasks; Round 33 proposals closed Sep 4, 2026	Outreachy Round 32
CNCF projects (Headlamp, Backstage, Kubeflow and others)	LFX	One project file per term; Term 3 2026 runs Sep–Nov	cncf/mentoring

Org	Program	Recent projects (2025–2026)	Ideas page
C4GT orgs (Glific, Avni, DIGIT and others)	C4GT DMP	Projects listed per ticket with scope and skills	C4GT portal · DMP 2027 guide
Summer of Bitcoin orgs	Summer of Bitcoin	2026: 26 orgs, 44 developer and 6 designer placements. Frontend-leaning orgs: Bitcoin Design, Nostr Components, Shopstr, Formstr, Jam, Cove. Designer track: wallet research, personas, high-fidelity wallet flows	2026 cohort
MLH Fellowship partners	MLH Fellowship	No public project list: you get matched to a project during the application. Over 100 projects a year; past frontend ones include React, Jest and DEV	Open Source track · Partners
ESoC projects	European Summer of Code	2026: pixi, pgmpy, pykalman, pytorch-forecasting, rattler, scikit-base, shap, sktime, skpro, OpenML, AI-on-Demand. Mostly Python and ML, little frontend	esoc2026 repo · esoc.dev

Not looked up yet: AOSSIE, Mattermost, Superset, Jenkins, p5.js, Sugar Labs, Mozilla, Mempool.space, BTCPay Server. Find them in the [GSoC 2026 org directory](#) or on each org's GitHub.

Skill roadmap

Learn by fixing real issues in your target org, not by finishing courses first. Each level unlocks a bigger class of issues.

Level	Skills	Unlocks
1. Base	HTML, CSS (Flexbox, Grid, responsive), modern JavaScript, Git (branch, rebase, resolve conflicts)	UI bugs, CSS fixes, docs-site issues
2. Framework	Your org's framework: React + hooks, Vue 3 Composition API, or Angular; TypeScript	Component fixes, small features
3. Project skills	Running a full stack locally (Docker, Node/Go/Python backends), state management (Redux, Zustand, Pinia, RxJS), REST/GraphQL	Features that touch APIs
4. Quality	Testing (Jest/Vitest, Cypress or Playwright), accessibility (WCAG, ARIA, axe), Storybook	Test-coverage and a11y issues mentors value
5. Senior signals	Performance (Lighthouse, bundle size), design-system work, reviewing others' PRs	Proposal-sized projects

Contribution playbook

The goal is for mentors to recognise your name before applications open. Follow these steps in order.

1. **Shortlist (week 1).** Check past program org lists for your stack. Keep orgs with recent merged PRs, responsive maintainers (replies within ~3 days) and `good first issue` labels.
2. **Set up (week 1-2).** Join the org's chat (Slack, Zulip, Discord, Matrix). Read CONTRIBUTING.md and the code of conduct. Run the project locally. Write down every setup snag you hit: fixing those docs makes a good first PR.
3. **First PR (week 2-3).** Ask to be assigned before starting, if the org works that way. Keep the PR small: one bug, one file area. Add before/after screenshots or a GIF for any UI change.
4. **Climb (month 2-4).** Move from fixes to small features, tests and accessibility work. Review other people's PRs and answer newcomer questions.
5. **Become known (month 3+).** Attend community calls. Share a short write-up or demo of what you built. Ask mentors which project ideas need doing.

6. Propose (Feb–Apr). Pick a listed idea where you've already touched that code. Share a draft proposal with the mentor 1–2 weeks before the deadline.

Etiquette:

- Never ping maintainers to merge or review. Wait 3–5 days, then follow up once.
- Don't claim more than one issue at a time.
- Don't submit AI-generated PRs you can't explain. Maintainers spot them, and it gets people blacklisted.
- Ask questions in public channels, not DMs, and include what you've already tried.

Proposal template

A strong frontend proposal shows the UI before you build it and proves you already know the codebase. Use this structure and adjust it to the org's own template if it has one.

```
# <Project title> – <Org>

## About me
Name, university/role, timezone, contact, GitHub. Two lines on your frontend background.

## My contributions to <Org>
| PR | What it changed | Status |
|---|---|---|
| #123 | Fixed dashboard layout on mobile | Merged |

## Problem
What is broken or missing today, with screenshots of the current UI.

## Proposed solution
- Mockups (Figma link) of the new UI
- Component tree and state flow
- APIs/data it uses; backend changes needed, if any
- Accessibility, responsiveness and testing plan (unit + e2e)

## Timeline (12 or 22 weeks)
| Week | Deliverable |
|---|---|
| Community bonding | Finalise design with mentors, set up Storybook |
| 1-2 | ... |
| Midterm | Demo-able feature behind a flag |
| Final | Merged, documented, tested |

## Stretch goals
```

```
## Availability
Hours per week, exams, other commitments.

## Why me / why this project
```

Tracker

Tick off these milestones as you go, and log every PR in the table so it's ready to paste into a proposal.

- ☐ Pick primary + backup org (by Oct 20)
- ☐ Project running locally, joined the community chat
- ☐ First PR opened
- ☐ First PR merged
- ☐ 3 merged PRs in primary org
- ☐ Attended a community call / introduced yourself to mentors
- ☐ 5 merged PRs, including one feature or test PR
- ☐ Outreachy initial essay submitted (Jan-Feb round)
- ☐ LFX Spring applications submitted
- ☐ GSoC org list checked (late Feb)
- ☐ Proposal draft shared with mentor
- ☐ GSoC / C4GT / Summer of Bitcoin applications submitted

Date	Org	PR / issue link	What it did	Status	Program it supports